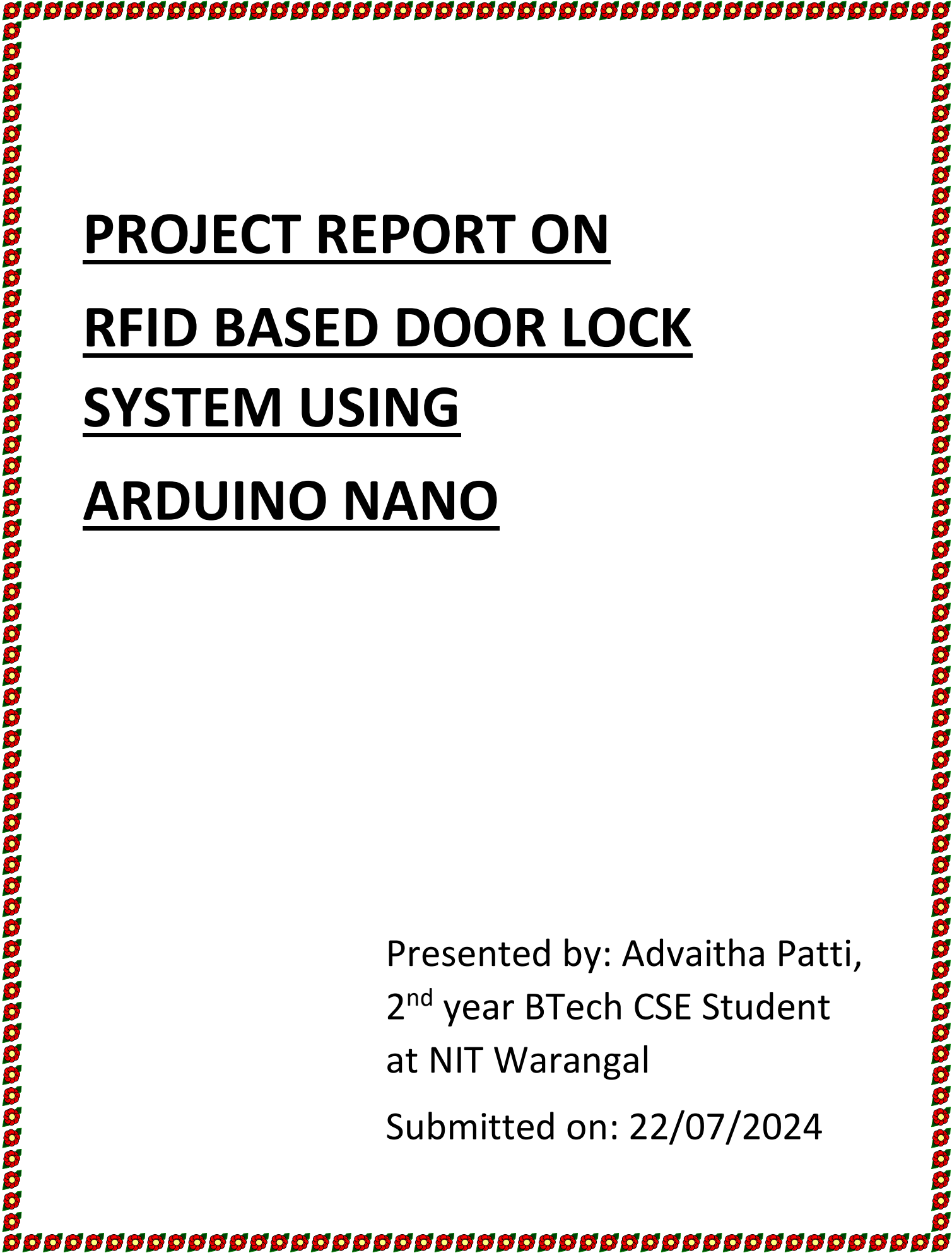




PROJECT REPORT ON **RFID BASED DOOR LOCK** **SYSTEM USING** **ARDUINO NANO**

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Introduction

With the increasing need for security in both residential and commercial areas, traditional lock-and-key mechanisms are becoming outdated. RFID (Radio Frequency Identification) technology offers a more secure and convenient alternative. This project aims to develop a door lock system that uses RFID technology to grant access to authorized personnel.

RFID(Radio Frequency Identification) technology is one of the leading technologies in the field of tracking things. Its applications are wide ranging from basic inventory management to Baggage tracking in the aviation industry.

RFID system consists of two main components namely RFID tags and readers.

The type of RFID tags used in the project are passive tags and it operates on HF(High frequency range).

The main advantages of RFID technology are as follows:

- Efficiency: RFID systems can read multiple tags at the same time, reducing the time required for data collection and processing.
- Accuracy: RFID reduces human error associated with manual data entry, providing more accurate and reliable data.
- Security: RFID tags can be encrypted, adding a layer of security and making it difficult to tamper them.

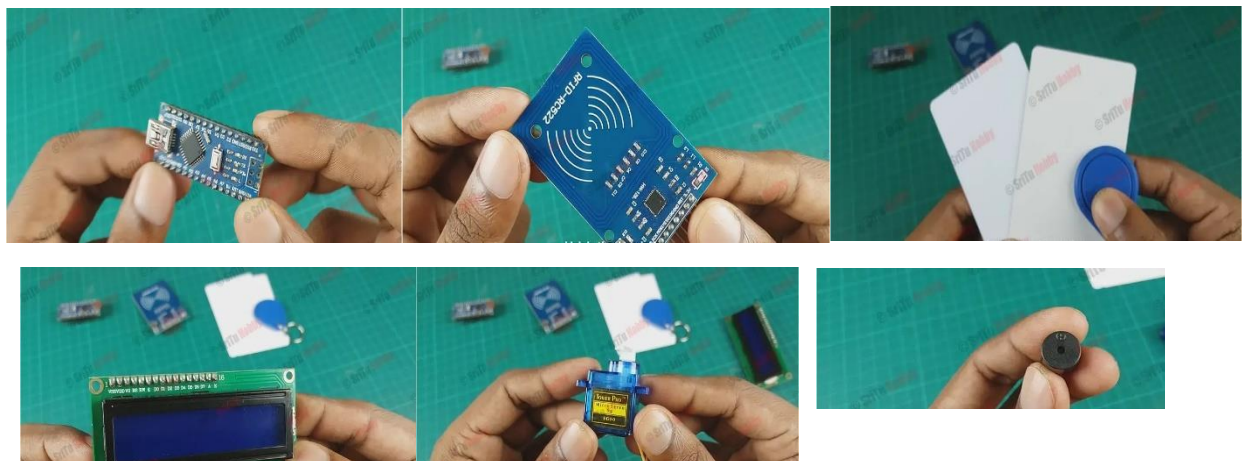
Objective:

- To design and implement a secure door locking system using RFID technology

Components and Materials:

Hardware Components:

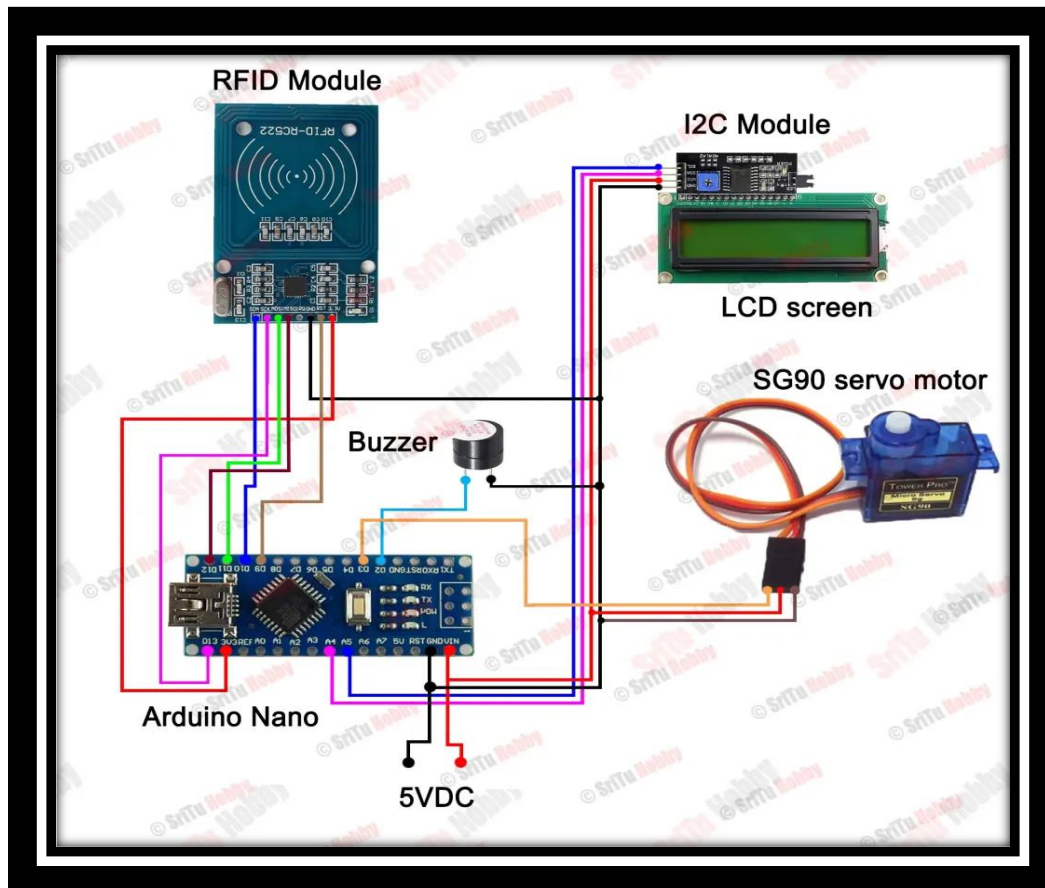
- Arduino Nano: Microcontroller board used for processing and control
- RFID Reader: Device used to read RFID tags
- RFID Tags: Unique identifiers for users
- LCD Screen(I2C Module): To display the UID of the tag scanned
- Servo Motor: Controls the locking mechanism
- Buzzer: Indicator for access status
- Connecting wires: For circuit connections
- Power source: for powering the Arduino
- Door locker
- Door hinges

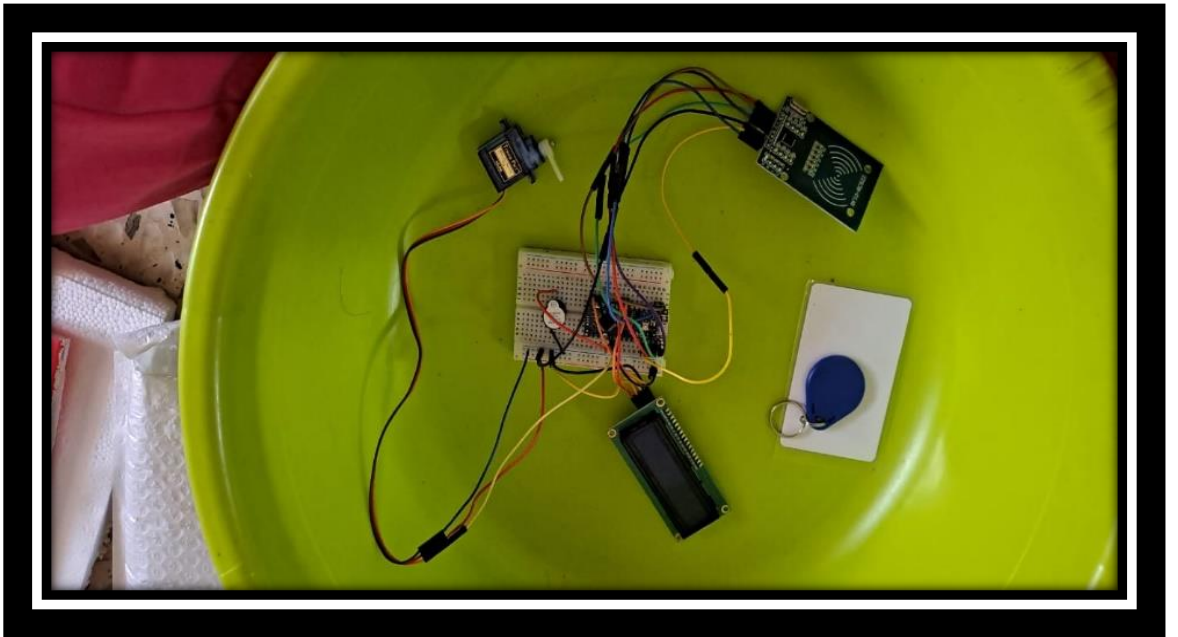
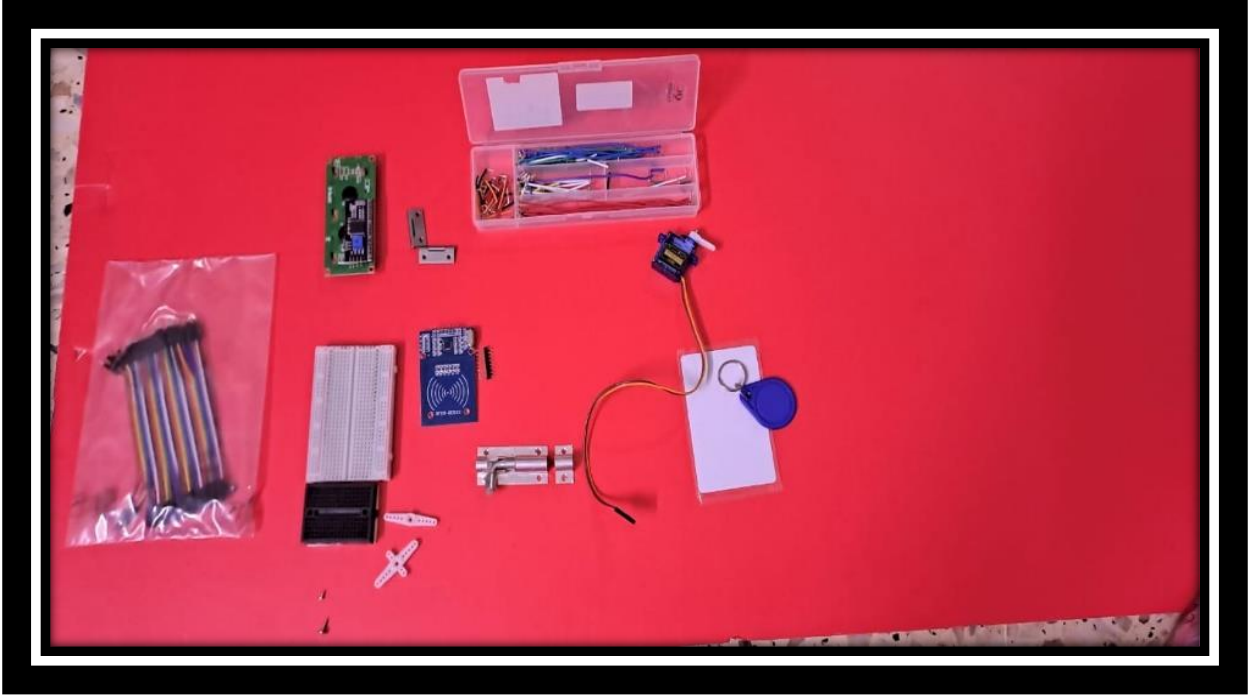


Software Components:

- Arduino IDE: Development environment for writing and uploading code to Arduino Nano
- MFRC522 Library: Library for interfacing with the RFID Reader
- Servo Library: Library for controlling the servo motor

CIRCUIT CONNECTIONS:





SOFTWARE IMPLEMENTATION:

Arduino Code:

1) UID Scan Code:

```
#include <SPI.h>
#include <MFRC522.h>

int rst_pin=9;
int ss_pin =10;

MFRC522 mfrc522(ss_pin,rst_pin);

void setup()
{
    Serial.begin(9600);
    while (!Serial);
    SPI.begin();
    mfrc522.PCD_Init(); // Initialising the RFID Reader
    delay(10);
    Serial.println(F("Scan the RFID card to see the
Unique ID..."));
}

void loop() {
    if ( ! mfrc522.PICC_IsNewCardPresent()) // Checks
if the scanned card is a new card or not
    {
        return;
    }
}
```



```
    if ( ! mfrc522.PICC_ReadCardSerial()) // Checks if
the scanned card ID is readable or not
    {
        return;
    }

    mfrc522.PICC_DumpToSerial(&(mfrc522.uid));
}
```

2) Main Code:

```
#include <Servo.h>
#include <MFRC522.h>
#include <SPI.h>
#include <LiquidCrystal_I2C.h>

int buzzer =2;
int servopin=3;
int rstpin=9;
int sspin=10;
byte lock;

String valid_id1="62 3D DD 51";
String valid_id2="B8 51 FA 12";

int delay1=200;
int delay2=1500;

MFRC522 rfid(sspin,rstpin);
LiquidCrystal_I2C lcd(0x27,16,2);
Servo servo;

void setup()
{
    SPI.begin();
    Serial.begin(9600);
    lcd.init();
    lcd.backlight();
    servo.attach(servopin);
    //servo.write(70);
    rfid.PCD_Init();
    pinMode(buzzer,OUTPUT);
}
```

```
}

void loop()
{
    lcd.setCursor(4,0);
    lcd.print("Welcome!");
    lcd.setCursor(0,1);
    if(lock==1)
    {
        lcd.print("Scan to unlock door...");
    }
    else if(lock==0)
    {
        lcd.print("Scan to lock door...");
    }

    delay(delay2);
    lcd.clear();

    if(!rfid.PICC_IsNewCardPresent())
    {
        return;
    }
    if(!rfid.PICC_ReadCardSerial())
    {
        return;
    }

    String UID="";
    for(int i=0;i<rfid.uid.size;i++)
    {
        UID.concat(String(rfid.uid.uidByte[i]< 0x10 ? "
0":" " "));
    }
}
```

```
        UID.concat(String(rfid.uid.uidByte[i],HEX));
        delay(300);
    }
    UID.toUpperCase();

    if((UID.substring(1)==valid_id1||UID.substring(1)==
valid_id2) && lock==0)
    {

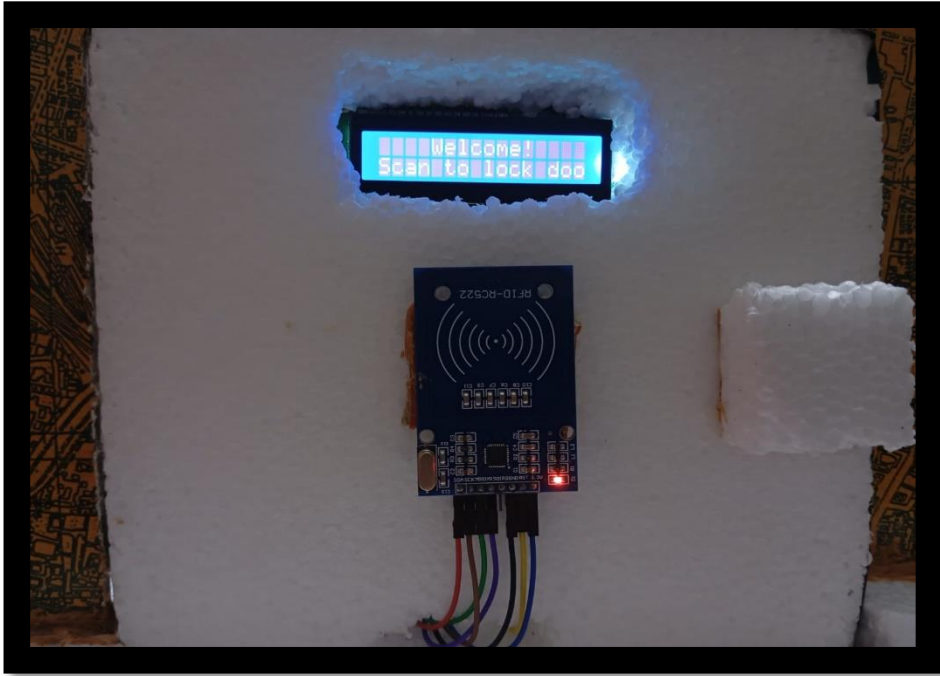
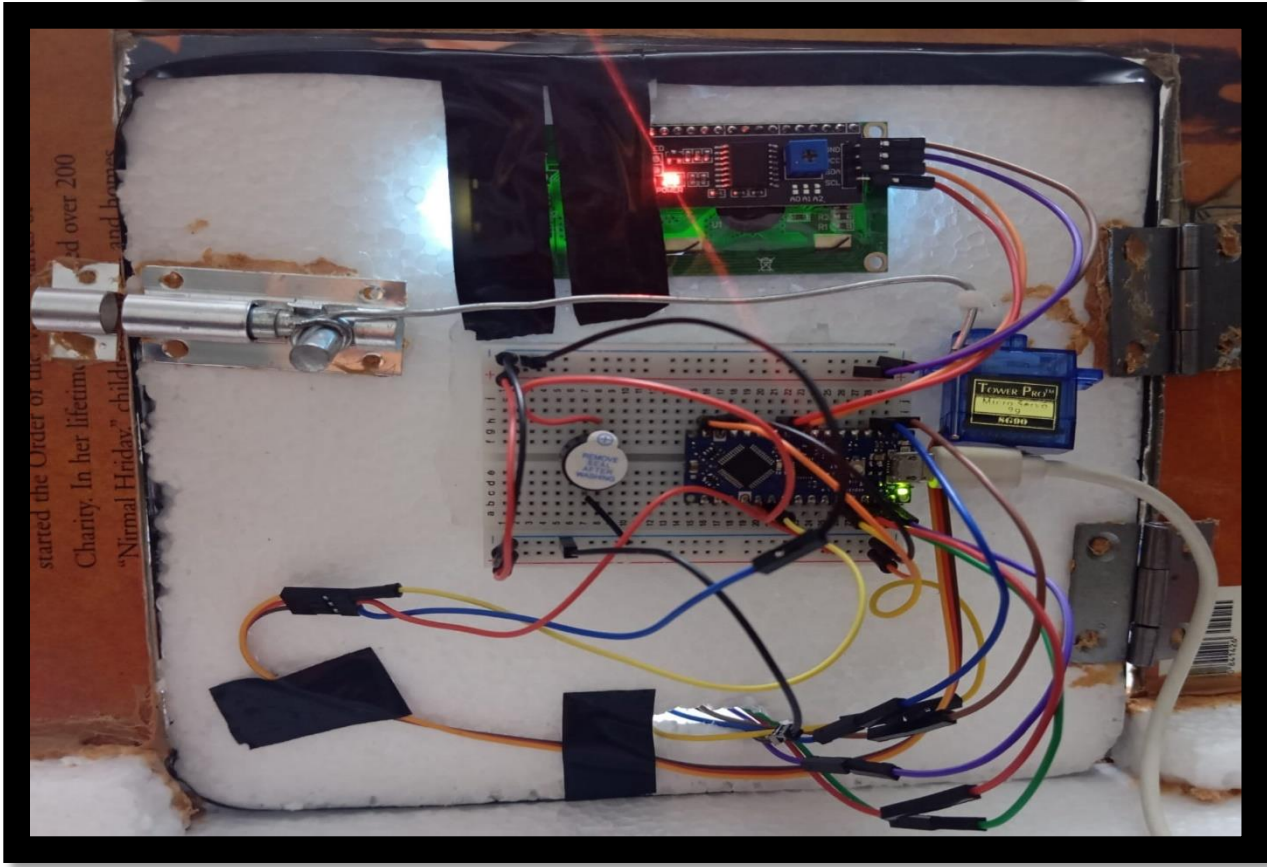
        digitalWrite(buzzer,HIGH);
        delay(50);
        digitalWrite(buzzer,LOW);
        delay(100);
        servo.write(50);
        lcd.setCursor(0,0);
        lcd.print("Valid Card...");
        lcd.setCursor(0,1);
        lcd.print("Door Locked..");
        delay(1500);
        lcd.clear();
        lock=1;
    }

    else
    if((UID.substring(1)==valid_id1||UID.substring(1)==va
lid_id2) && lock==1)
    {
        digitalWrite(buzzer,HIGH);
        delay(50);
        digitalWrite(buzzer,LOW);
        delay(100);
```

```
servo.write(110);  
lcd.setCursor(0,0);  
lcd.print("Valid Card...");  
lcd.setCursor(0,1);  
lcd.print("Door Unlocked...");  
delay(1500);  
lcd.clear();  
lock=0;  
}  
  
else  
{  
    digitalWrite(buzzer,HIGH);  
    delay(120);  
    digitalWrite(buzzer,LOW);  
    delay(120);  
    digitalWrite(buzzer,HIGH);  
    delay(120);  
    digitalWrite(buzzer,LOW);  
    delay(120);  
    lcd.setCursor(0,0);  
    lcd.print("Invalid Card!!!");  
    lcd.setCursor(0,1);  
    lcd.print("Access Denied!!");  
    delay(delay2);  
    lcd.clear();  
}  
}
```


WORKING:

- System Initialization: When the system is powered on, the Arduino Nano initialises the RFID Reader and other necessary components including the servo motor.
- RFID tag detection: The RFID reader continuously scans for the RFID tags in the nearby. When the tag is brought near to the RFID reader, the tag is detected.
- Reading the RFID tag: Upon detecting a tag, the RFID reader reads the unique identifier (UID) stored on the tag. This UID is a unique string of characters that distinguishes one tag from another.
- UID Authentication: The Arduino Nano compares the received UID against a pre-defined list of authorized UIDs stored in its memory. This list contains all the UIDs of the tags capable of unlocking the door.
- Access Control Decision:
 - ✓ If the UID is authorized: The Arduino activates the servo motor to unlock the door.
 - ✓ If the UID is unauthorized: The Arduino keeps the door locked and the buzzer buzzes two times indicating the denial of access.





RESULTS:

The system successfully reads RFID tags, authenticates them and controls the locking mechanism based on the tags' authorization status. Authorized tags unlock the door whereas unauthorized tags keep the door locked.

DRAWBACKS:

- Power dependency: The system relies on continuous power supply to function. In the event of a power outage, the lock may fail to operate unless a backup power source is provided.
- Limited Scalability: The current implementation stores authorized tag IDs directly in the Arduino's memory, which limits the number of users. For larger applications, an external database could be used.
- Physical limitations: The use of a servo motor may not be suitable for all types of doors or locking mechanisms.

CONCLUSION:

The RFID-based door lock system provides a seamless and secure method for controlling access to restricted areas. The use of RFID technology ensures that only individuals with authorized tags can unlock the door, while the Arduino Nano efficiently manages the authentication process and controls the locking mechanism. This system combines security, convenience, and simplicity, making it an ideal solution for various access control applications.